Exp 4
30/1/26

Packet capture tool : Wireshark

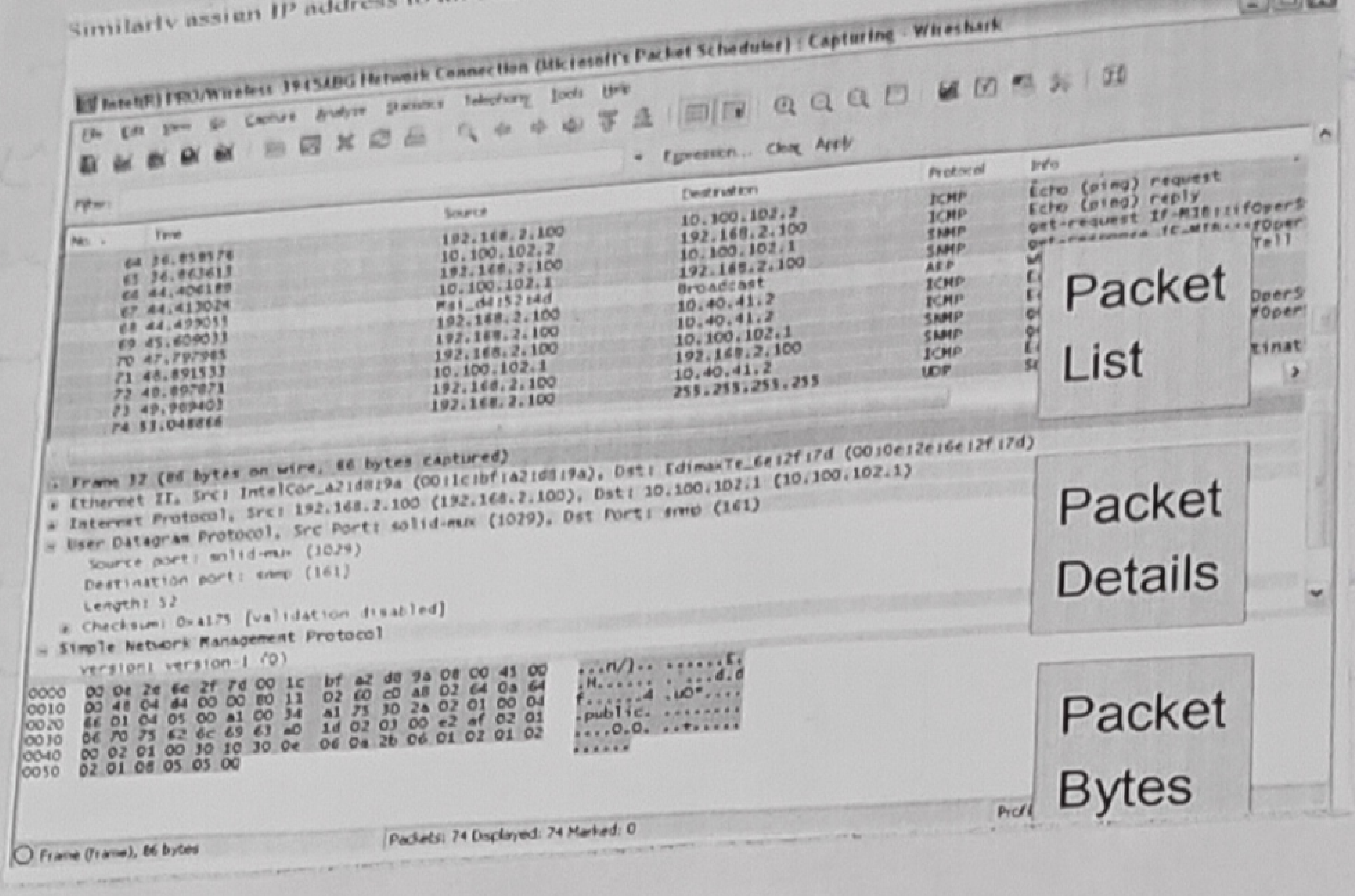
Aim :-

To capture, filter and analyze network packets using the wireshark packet sniffer and to observe TCP and UDP traffic along with the communication flow graph.

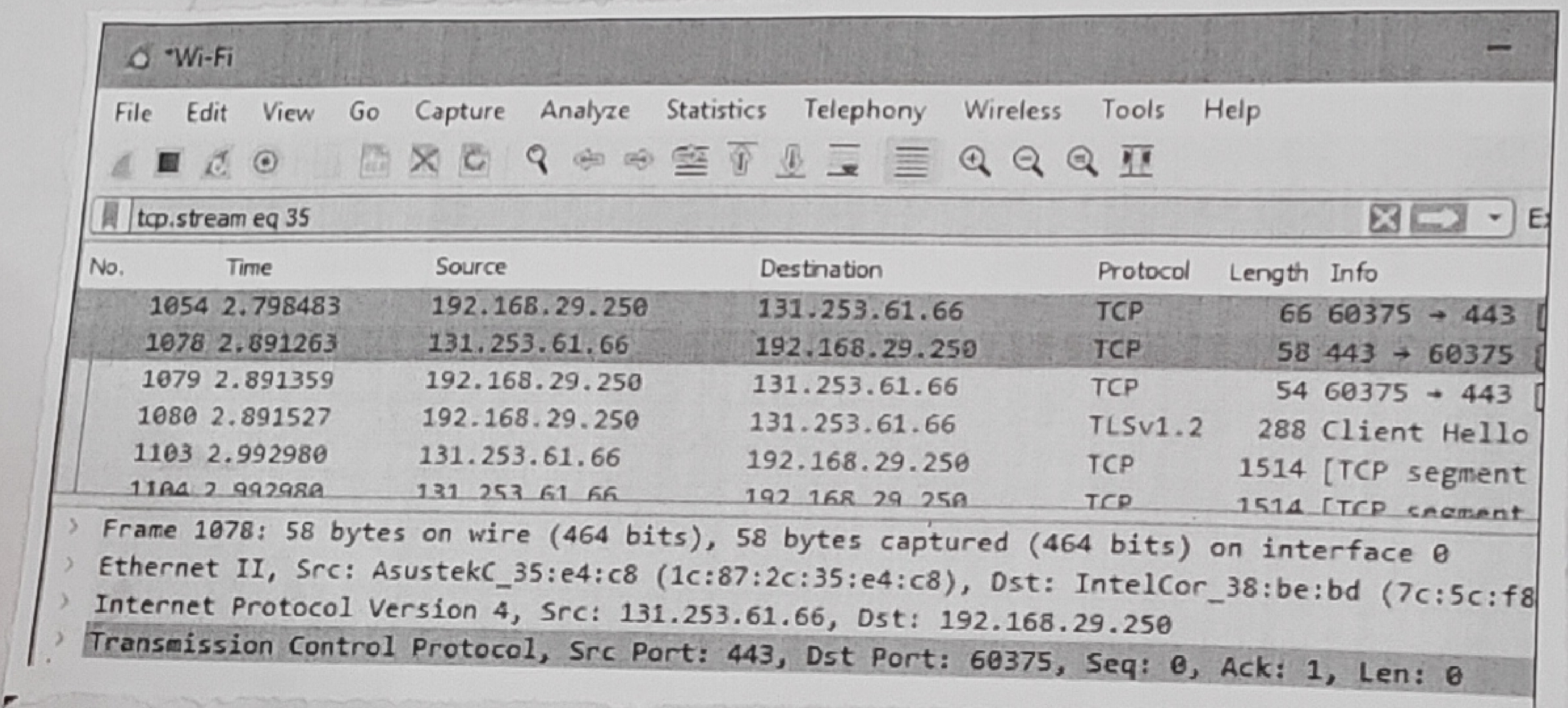
Procedure :-

- Open wireshark from the desktop or start menu.
- select the active Ethernet / wifi network interface.
- Go to capture → options
- Enable stop capture automatically after 100 packets
- click start to begin capturing packets
- Generate some network activity by opening a website or using internet applications.
- After 100 packets are captured, click stop
- In the Display filter box, type tcp or udp and press enter

Similarly assign IP address to all the PCS connected to switch.

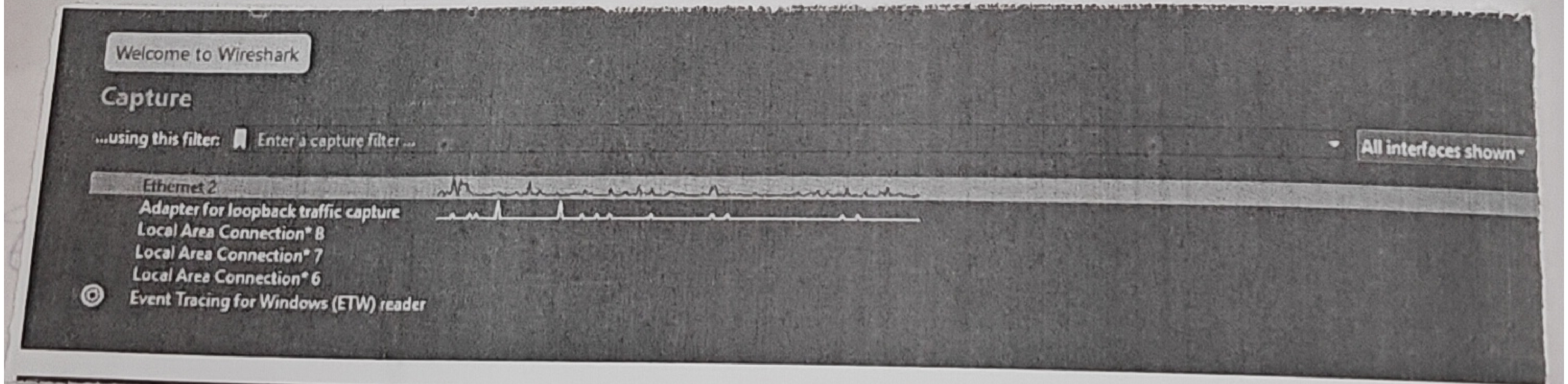

Intel(R) PRO/Wireless 3945ABG Network Connection (Microsoft's Packet Scheduler) : Capturing - Wireshark
File Edit View Go Capture Analyze Statistics Telephony Tools Help
Filter:
Expression... Clear Apply
Packet List
Packet Details
Packet Bytes

```
No. .   Time          Source            Destination        Protocol  Info
                      192.168.2.100     10.100.102.2       ICMP      Echo (ping) request
  64 36.858576        10.100.102.2      192.168.2.100      ICMP      Echo (ping) reply
  65 36.863613        192.168.2.100     10.100.102.1       SNMP      get-request IF-MIB::ifOperS
  66 44.406189        10.100.102.1      192.168.2.100      SNMP      get-response IF-MIB::ifOper
  67 44.413024        Msi_d4:52:4d      Broadcast          ARP       Tell
  68 44.499035        192.168.2.100     10.40.41.2         ICMP
  69 45.609033        192.168.2.100     10.40.41.2         ICMP
  70 47.797985        192.168.2.100     10.100.102.1       SNMP                                   OperS
  71 48.891533        10.100.102.1      192.168.2.100      SNMP                                   fOper
  72 48.897071        192.168.2.100     10.40.41.2         ICMP                                   tinat
  73 49.989403        192.168.2.100     255.255.255.255    UDP
  74 53.048866

- Frame 32 (86 bytes on wire, 86 bytes captured)
+ Ethernet II, Src: IntelCor_a2:d8:9a (00:1c:bf:a2:d8:9a), Dst: EdimaxTe_6e:2f:7d (00:0e:2e:6e:2f:7d)
+ Internet Protocol, Src: 192.168.2.100 (192.168.2.100), Dst: 10.100.102.1 (10.100.102.1)
- User Datagram Protocol, Src Port: solid-mux (1029), Dst Port: snmp (161)
    Source port: solid-mux (1029)
    Destination port: snmp (161)
    Length: 52
  + Checksum: 0x4175 [validation disabled]
- Simple Network Management Protocol
    version: version-1 (0)
0000  00 0e 2e 6e 2f 7d 00 1c  bf a2 d8 9a 08 00 45 00   ...n/}.. ......E.
0010  00 48 04 d4 00 00 80 11  02 60 c0 a8 02 64 0a 64   .H...... .`...d.d
0020  66 01 04 05 00 a1 00 34  a1 75 30 2a 02 01 00 04   f......4 .u0*....
0030  06 70 75 62 6c 69 63 a0  1d 02 03 00 e2 af 02 01   .public. ........
0040  00 02 01 00 30 10 30 0e  06 0a 2b 06 01 02 01 02   ....0.0. ..+.....
0050  02 01 08 05 05 00                                  ......

Frame (frame), 86 bytes        Packets: 74 Displayed: 74 Marked: 0
```

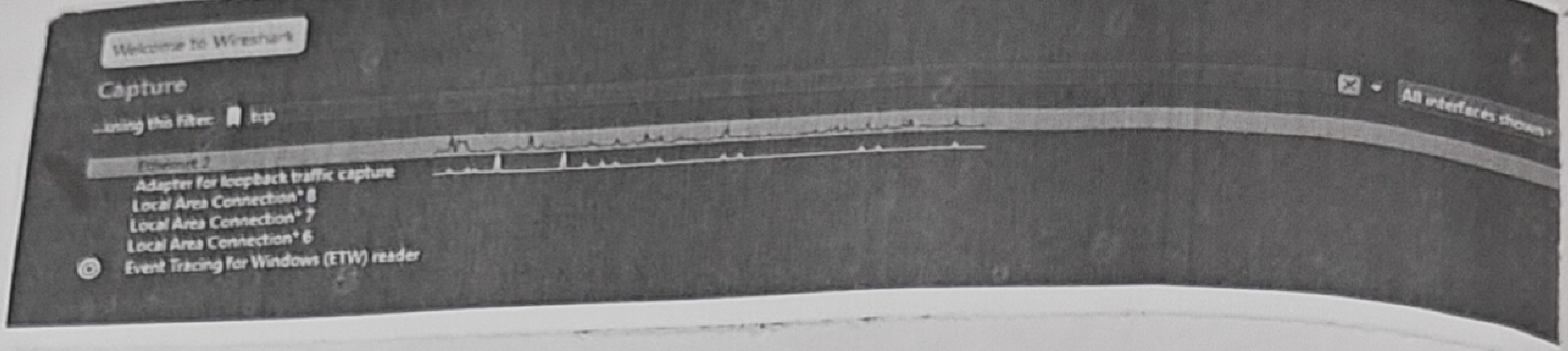

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 35

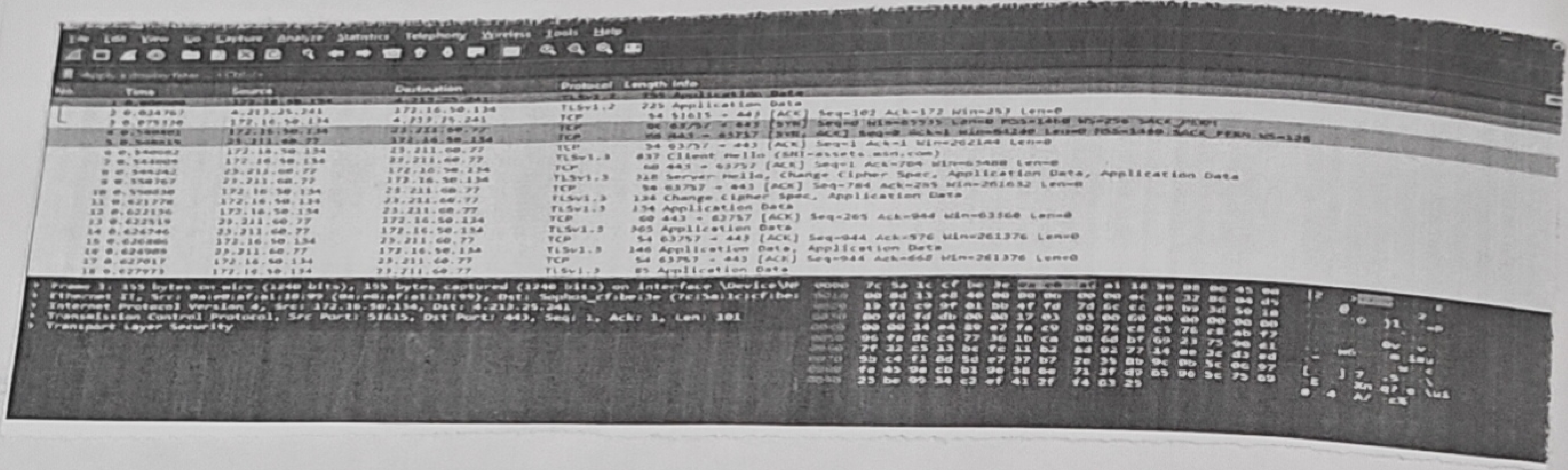
```
No.   Time          Source            Destination       Protocol  Length  Info
 1054 2.798483      192.168.29.250    131.253.61.66     TCP         66 60375 → 443 [
 1078 2.891263      131.253.61.66     192.168.29.250    TCP         58 443 → 60375 [
 1079 2.891359      192.168.29.250    131.253.61.66     TCP         54 60375 → 443 [
 1080 2.891527      192.168.29.250    131.253.61.66     TLSv1.2    288 Client Hello
 1103 2.992980      131.253.61.66     192.168.29.250    TCP       1514 [TCP segment
 1104 2.992980      131.253.61.66     192.168.29.250    TCP       1514 [TCP segment

> Frame 1078: 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface 0
> Ethernet II, Src: AsustekC_35:e4:c8 (1c:87:2c:35:e4:c8), Dst: IntelCor_38:be:bd (7c:5c:f8
> Internet Protocol Version 4, Src: 131.253.61.66, Dst: 192.168.29.250
> Transmission Control Protocol, Src Port: 443, Dst Port: 60375, Seq: 0, Ack: 1, Len: 0
```



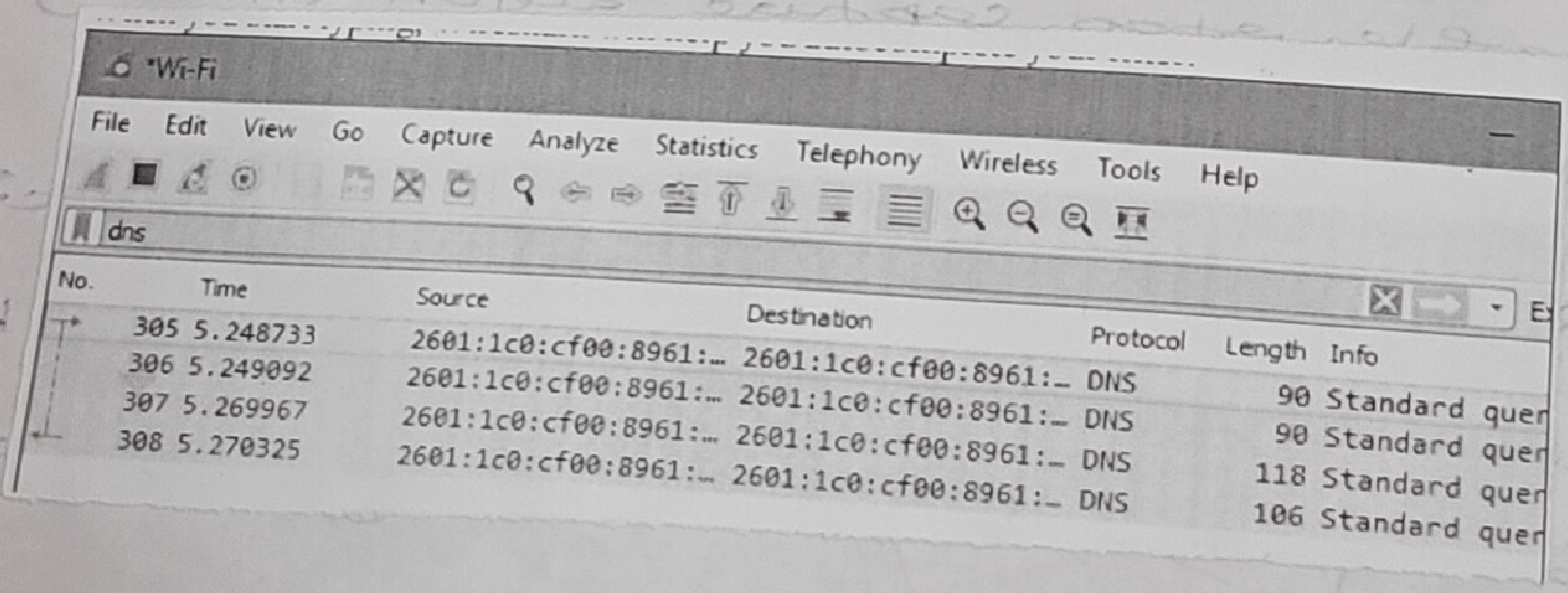
- Ethernet 2
- Adapter for loopback traffic capture
- Local Area Connection* 8
- Local Area Connection* 7
- Local Area Connection* 6
- Event Tracing for Windows (ETW) reader



Filter: tcp



File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info



*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Filter: dns

```
No.     Time          Source                 Destination            Protocol  Length  Info
  305 5.248733  2601:1c0:cf00:8961:…  2601:1c0:cf00:8961:…  DNS          90  Standard quer
  306 5.249092  2601:1c0:cf00:8961:…  2601:1c0:cf00:8961:…  DNS          90  Standard quer
  307 5.269967  2601:1c0:cf00:8961:…  2601:1c0:cf00:8961:…  DNS         118  Standard quer
  308 5.270325  2601:1c0:cf00:8961:…  2601:1c0:cf00:8961:…  DNS         106  Standard quer
```

- Observe the filtered TCP and UDP packets in the packet list pane.
- To view the communication flow, select statistics → flow graph
- Save the captured packets using file → save as.

Result:

Successfully captured 100 network packets using wireshark. TCP and UDP packets were successfully captured, filtered and analyzed using wiresha